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Status report

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Governance

Strategies, plans and policies

Communications minister Solly Malatsi withdrew the Draft National AI Policy in April 2026 after it was found to contain fictitious sources — AI-hallucinated references — in its reference list, and no national AI policy framework is in force: the redraft has not been gazetted, leaving AI already in use in hiring, lending and public administration ungoverned by any national instrument (July 2026). Malatsi convened an expert panel to redo the work, and the Department of Communications and Digital Technologies holds delivery: it says the rewrite will prescribe nothing to industry, issuing national guidelines only, with sectors writing their own AI strategies. The seven institutions the withdrawn draft proposed, a National AI Commission and a National AI Safety Institute among them, are being reconsidered, against a parliamentary committee preference for the opposite design: strengthening ICASA, the Information Regulator and the Competition Commission through the ICT and Media Regulators Forum launched in 2024.

Everything else runs through the Presidency and National Treasury. Cabinet approved the Digital Transformation of Government Roadmap on 26 March 2025, naming interdepartmental coordination and outdated systems among the problems it addresses, at the same meeting that approved Operation Vulindlela Phase 2, and the Roadmap organises the programme into four initiatives with dated milestones; the communications minister said on 21 September 2026 that its first phase closes in February 2027 and that government digital systems remain fragmented. Operation Vulindlela's own Q1 2026/27 report concedes that progress "has been uneven and, in certain instances, slower than initially anticipated due to the complexity of institutional reforms". A 2025 public service directive requires every department to set up a data governance committee and appoint a chief data officer.

Older commitments have aged without replacement. The National Integrated ICT Policy White Paper of 2016 still sets policy on universal access, e-government, digital identification and cybersecurity, and the 2007 free-and-open-source-software policy remains largely unimplemented inside the state's own archives (July 2026). The merger of Sentech and Broadband Infraco into a State Digital Infrastructure Company has slipped to 2028/29, more than a decade after it was first proposed, and the communications department has formally begun a structural review of what the State Information Technology Agency should be responsible for, the minister said in September 2026.

On price the department has conceded its own instrument failed. Four years after the 2022 spectrum auction, it records that expectations of lower network costs and consumer prices "remain largely unmet" and concedes a "current policy vacuum" (August 2026), and the study it has commissioned puts its own regulators under assessment. The deputy minister said in October 2026 that the redrafted AI policy is due for ministerial input before the end of November 2026 and at cabinet by the end of the 2026/27 financial year. The regulator withdrew both notices of its inquiry into over-the-top services on 30 September 2026, to republish them later, and the minister approved the universal service fund's manual in October.

The universal service agency gazetted a discussion document on the 2010 determination, with three options and a lean to targeted revision, comments closing 30 October 2026. The consultation, reported 10 October 2026, creates no new funding, subsidy or licensing obligation.

Legislation and regulation

South Africa's two largest mobile operators are in court against a consumer rule not yet in force. ICASA's sixth amendment to the End-user and Subscriber Service Charter Regulations was gazetted as GN 7027 in Government Gazette 53991 on 23 January 2026 and comes into force on 23 January 2027, requiring depletion alerts at 50%, 80% and 100% of a data bundle, barring out-of-bundle charging unless the subscriber opts in, and making unused bundles roll over automatically at least once. MTN and Vodacom have filed for review to have the rules declared invalid (July 2026).

Spectrum rulemaking has gone the same way. ICASA gazetted final innovation spectrum rules making the lower 6 GHz band licence-exempt for wireless ISPs, Wi-Fi and private networks while locking mobile operators out of it (May 2026), and both operators filed High Court reviews against the exclusive-use spectrum amendments (August 2026). On satellite the door stayed shut: a ministerial policy direction of December 2025 would have let foreign operators substitute equity-equivalence investment for direct Black ownership, ICASA rejected it in May 2026, and it could not have overridden the Electronic Communications Act anyway.

Courts are reforming more than Parliament. The North Gauteng High Court struck down the rule barring social grant applicants from putting new evidence before SASSA on appeal, and held that the Constitution requires government to plan for and progressively increase the grant's value and its income threshold (January 2025); the state's appeal is opposed, and the Supreme Court of Appeal heard it on 25 August 2026 and reserved judgment pending further affidavits, with an artificial-intelligence governance organisation admitted as a friend of the court on the harms of algorithmic decision-making. The Western Cape High Court then held that regulations made under the Births and Deaths Registration Act are subordinate legislation, so orders invalidating them need no Constitutional Court confirmation.

Where regulation has reached the largest platforms it has come from competition law rather than communications law. The Competition Commission's Media and Digital Platforms Market Inquiry closed on 13 November 2025 with binding remedial actions under section 43D(1) of the Competition Act for Google, Meta, Microsoft, OpenAI, TikTok and X, on a survey finding that 77% of the population take social media as their main online source of news, against 18% search and 4% personalised feeds. Google's remedies commit up to ZAR 687 500 000 over their five-year term, through news-showcase payments, an AI innovation fund, a transformation fund for small, independent and community media, and training.

The holes sit where the estate has grown fastest. Streaming services delivered over the open internet fall outside ICASA licensing and the Broadcasting Act 4 of 1999. SAHPRA has never commenced product registration of medical devices under regulation 8 of the 2016 general regulations, a power gazetted a decade ago and never switched on, though its definition of a medical device expressly includes software. Personal data is governed separately, under POPIA.

Data protection

A full bench set aside the Information Regulator's enforcement and infringement notices against the Department of Basic Education over publication of matric results in December 2025, and the Regulator's application for leave to appeal was refused in June 2026, regulatory decisions being appealable to the High Court under section 97 of POPIA. The Regulator also issued its first enforcement notice of 2026 against Central Johannesburg TVET College, for unlawful processing and failure to notify a compromise, and obtained High Court confirmation of a fine against

Blouberg Local Municipality for unlawfully publishing an employee's personal information (April 2026). In August 2026 it served an enforcement notice on the South African Bureau of Standards over its 2024 ransomware attack, finding unlawful processing and giving it 90 days to revise policies, run personal-information impact assessments and implement security measures, and referred two direct-marketing matters to its enforcement committee under section 69 on unsolicited electronic communications — against more than 3,800 complaints in the previous year, about a tenth of them concerning direct marketing.

The statute is the Protection of Personal Information Act 4 of 2013, in full effect since 1 July 2021, setting eight conditions for lawful processing and nine data subject rights, binding public and private bodies alike. It permits cross-border transfer only on stated grounds — comparable protection at the destination, informed consent, or a POPIA-equivalent contract — and South Africa publishes no adequacy register, so the burden of proving a destination adequate sits on each responsible party, per transfer.

Breach duties are unusually wide. Section 22 requires notification of both the Regulator and affected data subjects with no materiality or risk threshold, through a mandatory e-portal since April 2025, and a misdirected email has been treated as a reportable compromise. 788 notifications were recorded in the first quarter of 2026, and the Regulator reported more than 8,000 security-compromise reports to date and 1,220 in the five months from April 2026, a rate its chairperson called "very alarming" and expects to pass 3,000 for the financial year (August 2026); its executive for POPIA attributes the compromises to inadequate security controls — employee error, weak passwords, malware, ransomware and phishing — and says assessments find the public sector putting less effort than the private sector into appropriate security measures. The Regulator said on 30 September 2026 that it had received no section 22 notification from Gauteng's e-Government department over the exposure of the database behind its e-Panic Button app, and would engage the department.

The regime stops short of the technologies now processing the data. POPIA's application to AI is unsettled: no AI-specific guidance, no sector codes of conduct for high-risk uses such as credit scoring or healthcare support, and no compliance pathway proportionate to small firms (mid-2026), and coordination between the Regulator, sector regulators and the communications department remains limited, with the absence of enforcement capacity driving the uncertainty. The state's own next identity system is drafted below that line: the draft regulations give no right to erasure of biometric data, rely on weak or implied consent, and provide no citizen visibility into who has accessed their records (May 2026). Three Regulator seats were put out to nomination, with nominations closing 31 August 2026. The consumer commission launched the national opt-out registry, with direct marketers registering first and consumers able to register blocks from May 2027.

Regional collaboration

Regional Integration is one of South Africa's weaker measures in the Ibrahim Index, at 46.0 out of 100 and 29th of 54 African states, despite its membership of the Southern African Development Community (2023). The payments estate reaches further than that score suggests: the instant payment rail settles across borders through the SADC Transactions Cleared on an Immediate Basis scheme, with live real-time corridors to Zambia, Zimbabwe and the Common Monetary Area countries of Namibia, Lesotho and Eswatini.

The physical links run the same way. The state wholesaler Broadband Infraco reaches the borders of all six land neighbours and serves as their gateway to international capacity (2019), a private terrestrial fibre route has run from Mombasa to Johannesburg through the DRC since May 2023, the METISS cable has linked Mauritius, Réunion and Madagascar to a landing near Durban since March 2021, and the Umoja route from Kenya to Johannesburg had its terrestrial path built by May 2024, with an Indian Ocean crossing to Australia to follow.

Spectrum policy is written to regional and international allocation rather than to national preference. The National Radio Frequency Plan 2026 re-bases the 8.3 kHz to 3 000 GHz allocation table on the ITU Radio Regulations 2024, keeps South Africa in ITU Region 1 with national variations permitted only on condition of not interfering with other member states, and aligns to AfriSAP and the SADC Frequency Allocation Plan.

The tightest external discipline on the digital estate is financial. South Africa exited the Financial Action Task Force grey list in October 2025 after being placed on it in February 2023, and a fresh mutual evaluation runs to October 2027 — the frame Treasury and the Reserve Bank give for their cross-border crypto drafting. South Africa has also signed the agreement establishing the China-led World AI Cooperation Organization, set up in Shanghai in mid-July 2026.

Standards

The agency that holds the standards mandate is the one under investigation. The State Information Technology Agency Act 88 of 1998 gives SITA a statutory mandate to set and certify interoperability standards between government information systems; a Public Service Commission investigation found more than R2 billion in irregular expenditure alongside systemic governance weakness, leadership instability and procurement failure (July 2026), with 25.2% procurement attrition, contracts stuck for more than 400 days and departments seeking exemption from using it, and the communications minister gave SITA 30 days to produce a recovery plan (July 2026).

The standards exist and bind unevenly. Compliance with the Minimum Interoperability Standards, MIOS version 6.0, which prescribe data formats and technical standards for exchanging data between government systems, is obligatory for national and provincial departments under regulation 97 of the Public Service Regulations, 2016, and only informative for local government — the tier that runs most counter services. The government interoperability framework is recorded as only partially used, with its terms unpublished (2025), and the remaining interoperability standards are dated to September 2026. Health is building its own test: a conformance laboratory set up with the CSIR was to begin testing patient information systems against the gazetted health interoperability standards in January 2026. Internationally, the national standards body is a participating member of the joint ISO/IEC information technology committee and its subcommittees for security, identification cards, biometrics and AI.

The most fully specified standards regime in the country governs medical devices rather than government systems. SAHPRA requires quality management to ISO 13485:2016, risk management to ISO 14971 and software lifecycle to IEC 62304, with post-market monitoring for drift, classifies AI-enabled devices on the IMDRF-aligned Class A to D scale and requires Class C and D applicants to show pre-market approval in Australia, Brazil, Canada, the EU, Japan or the United States, or WHO prequalification, and expects evidence generated locally: independent test datasets, real-world conditions, subgroup analysis across sexes, ages and ethnicities, and a direct question to any foreign-trained model of whether it performs equally well on local populations. It then names every category of evidence and fixes no threshold: no minimum local sample size, no required subgroup performance, no operating point.

Where no public standard exists, a company's own fills the space: TikTok's requirement that creators label realistic AI content, and its adoption of C2PA Content Credentials, is the only synthetic-media labelling regime operating in South Africa (July 2026).

Public debate and participation in policymaking

Public perception of freedom of speech in South Africa fell 19.2 points over 2014-2023, to 69.4 out of 100 and 14th of 54 African states, even as the index's measured media freedom score rose (2023). The measured position remains strong: the Rights sub-category ranks 2nd of 54 at 80.0, with Freedom of Expression and Belief at 92.6, and the Participation sub-category 3rd of 54 at 76.4, with Civil Society Space at 75.9, up 5.1 points over the decade.

The channels for having a say are statutory rather than digital. There is no online public participation platform: the index records 'No' against the indicator and against all five of its sub-measures (2025), and a right-to-information law is in force but the same index marks 'No' on both sub-measures covering whether the public can actually obtain records and ask for what is not already published. What works is the gazette and the committee room: Parliament's call for written submissions on the Electronic Communications Amendment Bill was extended by a month on 20 August 2026, to 16:00 on 21 September 2026, consultation on the Digital Identity Regulations has closed with comments under evaluation for possible incorporation, and nominations for Information Regulator seats will be published with CVs for public comment on candidate suitability before shortlisting.

Access to those channels is a skill that has to be taught, and community network operators have been trained specifically to file submissions on communications regulation (2026). Where the estate meets land and resources, objection is local: there is no national moratorium or binding resource condition on data-centre development, and civil society is arguing that growth this energy-hungry "necessitates a conversation about prioritising" (July 2026), while objectors to the Cape Town data-centre rezoning may appeal to the municipal Appeal Authority, whose ruling is the City's final administrative decision (August 2026).

At the other end of the state, the conversation runs one way. The Department of Social Development is running a continuing blitz of unannounced ministerial inspections of local SASSA offices, framed as fraud control, with no fraud finding, count or outcome stated, and the minister attributing beneficiary complaints to users not understanding the system (August 2026).

Finance

Financial sustainability

South Africa's state broadband programme has moved from building to upkeep: the 2026 Budget Review records the second phase of SA Connect as ending in 2025/26, leaving the department to keep 948 government facilities connected each year, and the broadband subprogramme that carries it is appropriated R285.1 million for 2026/27, against R1.89 billion appropriated for 2024/25, of which R734.8 million was spent. The 2026 estimates print no financing source against their digital lines (2026/27).

The digital lines read from the national estimates come to about R13.3 billion appropriated for 2026/27, against about R12.6 billion for 2025/26.

Money voted for the core systems is not all spent. Home Affairs spent R1,020.7 million of the R1,110.6 million adjusted for its transversal IT subprogramme in 2024/25, and R828.5 million of R1,157.4 million on status services. The lines that keep those systems running are on the department's own vote: its R685.9 million modernisation programme for 2026/27 pays for maintenance and support of the revenue engine, live capture, e-Home Affairs hosting and public key infrastructure, and for Microsoft, IBM, HP and Gemalto licences.

Cost recovery is in law for the population register. Regulations under the Identification Act, in force from 1 July 2025, charge private users R10 for each real-time verification and R1 for each batch field, and state bodies nothing, a price the department set to fund maintenance of the register.

The state has said it will not carry its digital entities alone. The communications minister told Parliament in May 2026 that full state ownership of the department's entities is no longer sustainable, that partial privatisation makes sense in some cases, and that private operators should use state-owned assets under commercial agreements.

MoUs and other agreements

The largest thing signed for South Africa's digital economy is not a contract: Afreximbank and the Industrial Development Corporation signed a three-year renewable memorandum of understanding in Cairo in July 2026 to originate deals together against an $8 billion envelope, with Afreximbank saying it will run PAPSS, its MANSA customer due-diligence repository and the Africa Trade Gateway across the pipeline. An envelope of that kind sets a target to lend against; it commits no capital.

Government has been signing with itself on firmer terms. SARS and Home Affairs revived a bilateral agreement in place since 2010 on 3 April 2025, alongside a four-party memorandum with the Border Management Authority and Government Printing Works that supplies the governance framework, with Government Printing Works taking the roles of Certification Authority and Digital Trust Centre. At municipal level the instrument is looser: Johannesburg announced a sensor-and-analytics smart-city memorandum in July 2026.

New investments

Almost all the money going into South Africa's digital estate is corporate capital, and the largest financiers are hyperscalers and mobile operators spending their own balance sheets rather than development lenders — the largest single commitment on the record is Amazon Web Services' plan to put a further R30.4 billion into cloud infrastructure by 2029.

Concessional finance is marginal. Grant money amounts to little more than the two tranches of Germany's Digital Skills for Jobs and Income programme — a EUR 9.6 million grant running 2020 to 2024 and a EUR 6.5 million second phase running to 2027 — and technical assistance to five USTDA feasibility studies between 2017 and 2023, none larger than about $2 million. The development institutions that do lend, lend commercially: the IFC has put a $100 million senior term loan (2021) and a rand-denominated loan of about $58 million (2024) into Liquid Intelligent Technologies' fibre and data centre build.

The largest sums that have actually changed hands went to the state: at the ICASA auction of March 2022 Vodacom paid R5.38 billion for 110 MHz, the largest purchase in it, and MTN R5.2 billion for 100 MHz. Everything since has gone into buildings — Teraco is funding a 40MW hall at its Isando campus from an R8 billion syndicated loan arranged through Absa (2024), and Equinix has committed $438 million (R7.5 billion) of its own balance sheet to Johannesburg and Cape Town capacity (2026) — and into towers, where Actis completed its R6.75 billion purchase of Telkom's Swiftnet portfolio in March 2025.

Chinese money has come as vendor-linked buyer's credit for Huawei equipment, lent to Telkom and Cell C.

ICT Infrastructure

Connectivity

South Africa's remaining coverage gap is small, rural and now precisely located: 4G networks reach 98% of the population, yet roughly 400,000 households still have no 4G or better (March 2026), 74% of households sit within 10km of a fibre node while 12.2% are more than 20km from one, and more than 6,700 of the country's roughly 15,000 public schools and more than 1,000 clinics lie between 10 and 25km from the nearest fibre. The map was built for a development-bank investment study rather than published by the state, and its underlying dataset is not public. Connecting every household at 100Mbps would need R108bn to R142bn of cumulative investment on a 2025 development-bank estimate, the regulator's chairperson said in October 2026.

Access itself is close to saturated: 124 million mobile connections in January 2025, equivalent to 193% of the population, and around 79% of the population online in 2025. The Ibrahim Index records Internet & Computers rising 25.4 points over 2014–2023 to 67.5 out of 100, the largest ten-year gain of any of South Africa's 96 governance indicators.

What is bought on that access is changing faster than what is built with it. Data took 62.4% of Telkom's group revenue in the quarter to 30 June 2026; its data subscribers grew 15.5% to 19.8 million on traffic up 19.6%, while prepaid ARPU stayed broadly flat at R59 and group capital expenditure fell 19.4% to R888m in the same quarter. Take-up lags the build: Openserve passed 26,541 homes and connected 26,023 in the quarter to 30 June 2026, lifting its connectivity rate to 53.9%, so barely half the homes it has passed buy a service. The retail terms are themselves in court, with the two largest operators seeking review of Icasa's new data-bundle rules.

Spectrum remains the binding scarcity. Of 1,155MHz for which assignment plans or licences exist, 794MHz was assigned as at July 2026, and the regulator's own roadmap puts the country's requirement anywhere between about 3,000MHz and above 14,000MHz depending on how much traffic fibre and Wi-Fi carry — the scale question is demand rather than supply. On 25 September 2026 the communications department announced the gazetted withdrawal of the 2019 policy reserving high-demand spectrum for a wireless open access network that was never licensed, leaving that spectrum for ICASA to licence with wholesale, universal-service and anti-concentration obligations retained. Spectrum is also the one thing the smallest operators ask for: access to spectrum is the single common demand the community network sector makes of Icasa (July 2026). The Competition Commission recommended on 30 September 2026 that the Competition Tribunal approve MTN's acquisition of the tower operator IHS Holding, on conditions including fair and non-discriminatory access to its infrastructure for other mobile operators. The satellite operator said it has applied for licences to bring a direct-to-device service to South Africa with mobile operators, probably from around 2028 (2026). The regulator's chairperson said the authority has developed a draft satellite licensing framework (2026). An operator completed a 5G-Advanced trial in the upper 6GHz band under a temporary licence, reaching 10.35 Gbps on 400MHz (October 2026); the band is not yet licensed commercially.

Inside the state the network is slower than its digital plans assume: the Auditor-General's 2025/26 audit found 2,076 of about 7,800 sites on the state IT agency's network running below 10Mbps and 294 government facilities still on copper (reported October 2026).

Data Storage

Mission-critical government systems run in two national data centres operated by the State Information Technology Agency at Centurion and Beta, both National Key Points and reached through 34 switching centres, the communications department told Parliament's communications committee in February 2026, but government data is also hosted outside the country and the hosting arrangements are not published (2025), though the state does run a single shared government cloud platform. Localisation is sectoral: data centre infrastructure used by government entities must sit inside the country, and no general localisation mandate applies beyond that. The commercial build-out is now under constitutional scrutiny of a kind the regulatory frameworks do not provide: the Human Rights Commission opened a stakeholder process in May 2026 under section 184 of the Constitution, inviting written submissions by 30 July on eleven themes — among them electricity demand and tariffs, water use and cooling, environmental and climate impact, electronic waste, land use and community participation, privacy and cybersecurity, business and human rights due diligence, and equitable access to digital infrastructure — to assess whether the frameworks governing data centre development meet constitutional obligations. The commission said

in September 2026 that it had received more than 250 submissions, and that the key issue emerging was the availability, consistency and transparency of information on electricity and water demand, land use, environmental impacts and impacts on surrounding communities; it has published no finding, timetable or report date.

The most prescriptive residency rule yet drafted sits in exchange control rather than data policy. The Reserve Bank's draft Crypto Asset Manual would let authorised providers host offshore within a group, but refuses any arrangement storing data in a sanctioned country or a jurisdiction that may inhibit effective access to it. Approved offshore hosting would have to be ring-fenced by contract, retained five years and produced to the Financial Surveillance Department within 48 hours, and a provider leaving the country would have to replicate five years of data back into it.

The estate that would hold any of this is foreign-owned and growing. South Africa has 68 data centre facilities (2025). Amazon Web Services opened its Cape Town region, its first in Africa, in April 2020, and Google's Johannesburg region became operational in January 2024. Microsoft put R20.4 billion into its Johannesburg and Cape Town regions over the three years to March 2025 and is adding R5.4 billion, about $329 million, for land, power and water readiness (April 2026); Equinix has bought R890 million of land, some 327,000 m² (April 2026). Siting is where the build meets resistance. Cape Town's Municipal Planning Tribunal approved the land use for two Equinix data centres by four votes to one in July 2026, but that approval is under formal appeal by two civil society organisations, who say the application lacked information on electricity, water, emissions and noise and that nearby communities were not consulted (September 2026), and the City confirms the rezoning is suspended and cannot be acted on until the appeal is finalised by the executive mayor as planning appeals authority, while the operator calls the site a long-term land bank on which it has submitted no planning or development application (September 2026). One of the appellants wrote that the appeal went before the City's Planning Appeals Advisory Panel on 12 October 2026, after which the mayor has 90 days to decide. The City is drafting new development guidelines for large data-centre applications, its first public acknowledgement that the approvals process needs changing. The communications department proposed in its February 2026 presentation a national integrated government data centre as a medium- to long-term measure, with government using industry capacity in the meantime, and stated no budget, site or date. The State Information Technology Agency's plan to offer departments public cloud across several hyperscalers was 55% implemented against a 100% target in 2024/25, with service level agreements still to be concluded, and it awarded the police service a five-year, R339 million Huawei cloud infrastructure contract by closed bid. Research computing sits apart: the National Integrated Cyber Infrastructure System, run by the CSIR across high-performance computing, the research network and DIRISA's secure data centre, whose Centre for High Performance Computing has reached maximum capacity (August 2026). A hosting provider for online gaming announced its first African data centre, in Johannesburg, on 30 September 2026, without disclosing the site or its capacity.

Energy

The constraint on South Africa's grid has moved from generation to distribution. Load shedding has largely stopped: the year-to-date energy availability factor reached 67.55% by August 2026, the strongest in six years, with unplanned outages nearly halved year on year, yet households across the provinces went on losing power through 2025, mostly through faults in municipal distribution networks rather than national supply shortfalls.

The supply behind that remains overwhelmingly coal-fired: wind, solar and other renewables supplied 17.8TWh of the 221.2TWh generated in 2024, with several thousand megawatts of private generation registered by mid-2024 after licensing thresholds were lifted to 100MW. Connection is near-universal — 87.7% of the population had electricity access in 2023 — and the urban-rural gap has reversed: rural access reached 94.0% against 90.9% urban in 2023, the backlog now sitting in urban informal settlements.

For digital load the live question is connection rather than supply. National Treasury has reclassified data centres as infrastructure as critical as electricity, ports and transport, signalling priority grid connections and tax incentives (February 2026), and Eskom's public position is that current data-centre demand is within what the grid can carry (July 2026). Operators are hedging regardless: Teraco's expanded 50MW Cape Town facility runs zero-water closed-loop cooling and a 120MW solar PV wheeling plant (November 2025), and its Johannesburg build wheels utility-scale solar across the Eskom network.

Nothing yet binds that build-out to a resource condition. Equinix's two planned Cape Town data centres, of up to 160MW, drew a formal objection that the application disclosed neither their water nor their power demand, in a city with the "Day Zero" drought in living memory (June 2026); the planning tribunal deferred both questions to a later stage (July 2026), and there is no national moratorium or binding resource condition on data-centre development. At the other end of the estate, rural primary healthcare clinics still record encounters on paper because connectivity and power are unreliable.

Technical Capacity

The state's central IT agency is rebuilding an executive that stood largely empty: its managing director said in September 2026 that 60% of executive committee posts had been vacant and that a permanent chief financial officer and executives for client engagement, supply chain, and network, cloud and platform have since been appointed. The Public Service Commission's investigation of government IT procurement over 2020 to 2025 named that as a root cause of the instability, finding 86% of managing-director roles held on an acting or interim basis, board tenures too short for strategic continuity, and irregular expenditure above R2bn across four audited years read as control failure through absent deterrence. Its managing director since April 2026, Magatho Mello, is the first permanent holder of the post since late 2023, appointed as the minister constituted a new board. It carries more than 100 departments and public entities, the police service and defence department among them, and some client departments are looking elsewhere. The same shortage surfaces wherever a department is obliged to write it down. The Border Management Authority has gone to market for three years of software-development resourcing, conceding in the tender that its internal capacity is insufficient to meet evolving system demands (July 2026), and the deputy minister of justice named ICT human resource capacity, procurement delays and budgetary pressure as what holds back his department's digital estate (July 2026).

What cannot be staffed is bought in. Home Affairs' own strategic plan describes the department as grossly underfunded and dependent on third parties for critical services, and the identity infrastructure it runs rests on foreign suppliers for both fingerprint matching and smart card production. The government data-exchange platform MzansiXchange was nine months into pilot and onboarding additional technical capacity to complete it (July 2026), and the Presidency's own reform programme concedes that digital progress has been uneven and slower than anticipated because of the complexity of institutional reform. Cyber security roles are largely unfilled.

There is a domestic base underneath the shortfall. South Africa's GitHub developer population grew by roughly a quarter over 2024, and a South African firm has opened a Midrand headquarters with a UPS test platform and a cooling test centre certified to European standards, supplying the power and cooling layer beneath the foreign-owned data-centre build (July 2026). The domain registry counted 1.5 million active registrations across its four commercial namespaces in October 2026.

Cybersecurity

South Africa's national CSIRT publishes nothing by which its own work could be judged: no statistics page, no annual or quarterly report, no advisories archive and no data downloads; the newest dated item on its site was from September 2025, and on 4 August 2026 the site served an HTTP 503 maintenance page. Neither the register of

sector-CSIRTs the Cybersecurity Hub is mandated to establish nor any national standard it is mandated to create has been published, so fourteen years of the National Cybersecurity Policy Framework, gazetted in December 2015 and still the country's policy instrument, cannot be assessed from anything the responsible body has put out. On the measure that scores commitments rather than outcomes, South Africa sits in the "advancing" tier of the ITU's Global Cybersecurity Index 2024, for states scoring at least 85 out of 100.

Fraud against operators and their customers is measured where intrusion is not: the industry body records 14,897 subscription fraud cases in the twelve months to April 2026, up 307% on the year before, with every month above its counterpart, social engineering over messaging and calls having overtaken technical intrusion and AI-generated voices used to carry it. A free helpline for victims of online fraud, hacking and digital abuse opened in October 2026 as a 12-month pilot, staffed by supervised students and funded by a US cloud provider. The incident record underneath is heavy. Of 309 organisations in the CSIR's 2024 national survey, 88% had been breached in the previous twelve months and 90% of those more than once, with third-party connections into the enterprise the leading root cause at 48%, ahead of end-user phishing at 45% and only 20% applying least privilege. Across 291 public-sector institutions surveyed in 2024, 47% had between one and five incidents and only 16% had none, while 64% rated themselves very prepared and 56% had suffered a data breach.

The departments building the state's most sensitive systems say the rest themselves. Home Affairs concedes its cybersecurity maturity remains low and that it lacks a fully implemented disaster-recovery strategy, with a security operations centre and new information-security posts planned but not in place (April 2026), while it builds the national identity rail. The Electoral Commission has readvertised a tender to rebuild its public mobile app, stating that the app runs on a framework no longer supported, poses security risks and has been found to have vulnerabilities and inadequate provisions for data safety and user permissions (July 2026). Gauteng's provincial e-Government department left the database behind its e-Panic Button app unsecured, exposing users' names, phone numbers, crime reports with images and GPS coordinates, location histories back to the app's 2024 launch and one-time login PINs, until its contractor fixed the flaws after a news outlet's alert on 21 September 2026. The provincial legislature's e-government committee rejected the department's assurance that no personal information was compromised and gave it seven days to report (October 2026). Statistics South Africa was hit by a ransomware attack in March 2026. The pathology group Ampath confirmed an attack detected on 2 October 2026 in which identity numbers and medical-aid details may have been reached, disputing an attacker's claim of 80 million records as a count of database entries rather than patients. Government elevated cybersecurity to a counter-intelligence priority in July 2025, placing protection of state systems inside the national intelligence mandate.

The legislature's committees said after a joint meeting on 5 October 2026 that the department notified the Information Regulator only that morning, about two weeks after it was alerted, and could not say how many records were exposed.

DPI

Data Exchange

South Africa's government data exchange was nine months into a one-year pilot and onboarding additional technical capacity to complete it as at 31 July 2026. The National Treasury launched MzansiXchange in October 2025, a federated platform built on Estonia's X-Road and designed as a real-time rail between institutions rather than a central repository. Its first live use case links Home Affairs data to the South African Social Security Agency for real-time verification, and is in testing rather than in full production.

Nothing whole-of-government sits behind it yet. Several partial systems run in parallel — the Treasury's Secure Data Facility, the social security agency's relief-of-distress system, the cooperative governance department's National Strategic Hub and the South African Integrated Data Lake — and the real-time exchange is still to be built, under an inter-departmental working group set up in September 2024. The core financial machinery is joined by older means: tax administration, customs, human resources and financial management are connected both through separate point-to-point interfaces and through a Government Service Bus. A government interoperability framework is in place but only partially used, and its terms are not published, alongside the minimum interoperability standards that bind national and provincial departments. The integrated social information system meant to join social protection data remained undeveloped in June 2025, with no legal framework for sharing it. Health runs its own exchange, which the health ministry says now carries the patient register and the facility list (June 2026).

The heaviest cross-government traffic runs outside the exchange altogether. The social security agency's grant verification cross-references records held by the revenue service, the Unemployment Insurance Fund, the Government Employees Pension Fund, banks and credit bureaus — a bilateral apparatus the courts have already cut into. In January 2025 the North Gauteng High Court declared unconstitutional and invalid both the relief grant's checks against databases that may indicate income and the proxy means test conducted by verifying insufficient means with banks, a ruling the state has taken to the Supreme Court of Appeal.

Digital Identity and CRVS

The only digital identity target Home Affairs set itself for 2026/27 is the hosting infrastructure, due complete by 31 March 2027, with verifiable credentials and digital wallets placed explicitly in later phases and full digital identity promised "by 2029". The minister showed the first working digital identity credential, bound to his own phone, on 1 October 2026, with no public release date given. The public key infrastructure and identity platform are being built inside the South African Revenue Service's hosting environment rather than Home Affairs' own, with facial recognition the primary biometric and fingerprints secondary.

The legal basis is not yet in place. No digital-identity legislation has been enacted, and in February 2026 President Ramaphosa confirmed the framework was still awaiting Cabinet approval. Draft Identification Regulations gazetted in May 2026 would create an explicitly optional smartphone credential, with accreditation of "trusted entities" and seven-year audit-log retention; consultation has closed and the comments are under evaluation, so the regulations are not made.

The card keeps moving. Home Affairs issued 4,002,964 Smart IDs in calendar 2025, 17% up on 2024. Smart ID services ran at more than 500 branches of Capitec, Standard Bank, FNB and Absa, with at least one in every province, as reported on 27 September 2026, and the minister says the channel will reach 750 branches and add first-time smart ID and passport applications and home delivery by the end of 2026. More than 500,000 people switched off the green barcoded book through bank branches in under five months, against roughly 16 million green books still in circulation (August 2026), and production of the green book ends on 31 March 2027, when its machines are to be decommissioned, the minister said on 1 October 2026, putting smart ID replacements at more than 515 bank branches and 822,000 users; no date on which existing books stop being valid is announced. 360,000 documents lay uncollected at Home Affairs offices as at 31 July 2026. 4,432,923 South Africans aged 16 or over had been issued neither a green ID book nor a smart ID card as at November 2025.

Biometrics reached the benefit counter well ahead of the wallet: enrolment has been required of every new grant application and of triggered reviews since 1 September 2025, and 7,779 complaints have been logged about the facial biometric system against none about fingerprint enrolment, the remedy offered being to travel to a local office.

Civil registration is the weak end. 83.2% of births in 2024 were registered within the legally mandated 30 days, but registration at a Home Affairs office is an in-person paper process on form DHA-24, and online birth registration had reached 242 health facilities by 2024/25, with the 2025/26 plan targeting only nine more. In July 2026 the Western Cape High Court declared the birth-registration regulations unconstitutional for four categories of exclusion, three of which Home Affairs conceded, with 18 months to amend the rules. The department's quarterly report records a missed first-quarter target to start procuring the hosting infrastructure, and another to revise the identification and registration Bill for public comment. The transport department told Parliament a Presidency-led working group is exploring digital driving licences within the government's digital identity framework (2026). The transport department briefed Parliament on draft regulations extending light-vehicle licence cards from five to ten years, citing breakdowns of the card printing machine (2026).

The transport department plans to amend the road traffic regulations so that traffic register number certificates, learner's licences and driving licences no longer count as identification, with a number valid only as long as the passport and permit behind it (October 2026); the minister said 300,000 dormant numbers had already been blocked.

Digital Payments and Fintech

Postbank had replaced 172,146 of the 463,836 active SASSA gold cards by 28 July 2026, leaving 264,263 outstanding, with the gold cards due to stop working after 31 August 2026 and black replacements to be issued free over the counter at Shoprite, Checkers, Usave, Pick n Pay, Boxer and Spar. Postbank places those sites using the location of each recipient's last three transactions, ring-fencing any customer more than 30km from one. Grants reached about 19 million people in 2026, the country's largest single payment-instrument base, almost entirely into bank accounts and onto cards.

The state has bought into the rails it regulates. The Reserve Bank took a 50% stake in PayInc, operator of the instant payment rail, and the February 2026 Budget recast the clearing house as open, shared national payment infrastructure. Governor Lesetja Kganyago said in August 2026 that the Bank would have preferred to hold all of PayInc and is not admitting non-bank shareholders, though 39 banks and more than 480 non-bank payment service providers sit on the rails. PayShap, the instant rail live since March 2023, processed more than 500 million transactions worth about R486bn in 2025 but had passed only six million registered users by May 2026, held back by owner-bank pricing of R1 to R10 a transaction and the absence of a USSD channel for feature phones.

Reform is running ahead of the statute. Activity-based licensing to let non-banks issue e-money and reach clearing and settlement directly is being delivered through an exemption notice and a forthcoming National Payment System Bill, a national interoperable QR standard has been gazetted with PayShap the first rail to adopt it, and a state payment credential and wallet are in active development, with testing against government use cases dated November 2026. The modernisation programme uses neither tokenisation nor a distributed ledger. The Reserve Bank has authorised a UK money-transfer firm as a category 2 foreign exchange dealer (October 2026).

Crypto has outrun that design. More than 300 crypto asset service providers are licensed, and dollar-stablecoin flows of about R27bn ran on-chain in the year to 30 April 2026 while licensed providers' monthly trading fell from about R19.2bn in 2025 to about R10.7bn over the first four months of 2026. Absa launched custody of digital assets for institutional and large business clients on 21 September 2026, not a retail offer.

Registries

377,060 identity numbers remained blocked on the National Population Register in July 2026, down from 702,267 when the Gauteng High Court found the placement of markers without due process unconstitutional; Home Affairs blames "unserviceable addresses" for failing to trace around 400,000 affected people, and the court deadline has

been extended to March 2027. That points at a second gap: there is no authoritative national register of addresses, and no commercial, statistical, municipal or postal address set holds a national mandate or complete coverage, leaving the companies commission to verify company addresses through a commercial geolocation service.

The population register is the identity key that the digital-ID programme, the revenue service and the grant and unemployment-insurance systems read from. Home Affairs raised a real-time verification from 15 cents to R10 per check, effective 1 July 2025 to fund an Intelligent Population Register, and the industry association for Vodacom and MTN has filed a High Court review calling it a regressive tax. The verification service failed up to half of requests in March 2025, before an upgrade brought its error rate below 1%. The President has cast that register as a migration-control instrument holding biometrics for every person in the country, with the green ID book to be de-recognised.

Registers are joined bilaterally. Companies registered through BizPortal are automatically registered with the revenue service, the Unemployment Insurance Fund and the Compensation Fund, with the applicant's identity verified in real time against the population register. The Electronic Deeds Registration System launched on 1 April 2025 as a controlled beta test, though its electronic registration module is still in pilot and no deed had been registered end to end on it by November 2025. Voter registration still depends on a voter presenting an identity document rather than on an automated link to the population register, and a draft bill would have Home Affairs pass records to the Electoral Commission at 16 and to the roll at 18; interoperability between the two is targeted only for the 2029 elections.

Errors stay with the citizen: identity numbers stolen to make Covid-era relief claims still show their owners as employed on the Unemployment Insurance Fund register, blocking those workers from benefits, and neither the labour department nor the police will take ownership of correcting the record. A beneficial-ownership disclosure module launched on 29 September 2026 gives the police, the prosecuting authority and other law enforcement agencies real-time access to company ownership information.

Health keeps its own key: the patient registration system runs in 3,280 facilities, with biometric registration verified in real time against the population register in 249 of them (August 2025). More than 49 million paper civic records had been digitised since 2023, the deputy home affairs minister said in July 2025.

Sectoral management information systems

The National Student Financial Aid Scheme was placed under administration in May 2026, citing delays in ICT modernisation and systems integration, and names ICT as its biggest operational risk — the second governance intervention on the same fault line, after its board was dissolved in 2024 over IT-system failures and the fintech disbursement debacle.

Health is the largest unfinished build. The unified electronic medical record had reached beta implementation in 36 primary healthcare facilities and one general practice by March 2025, with scale-up across primary healthcare facilities not due to begin until January 2026, and rural clinics remain paper-based at the point of care, clinicians recording encounters by hand and entering them into national systems such as DHIS2 and the patient registration system afterwards, constrained by unreliable connectivity and power outages. By June 2026 the state-built record was still in beta and user-acceptance testing, and the department's tender was for rollout and training rather than a new system. The strategy that was to have closed that gap ran to 2024 and committed the department to conformance testing of all health information systems, a Master Patient Index carrying a unique identifier across provinces and a roadmap to an electronic health record, together with a national digital health grant targeting three to five per cent of health expenditure. The ministry promised a revised strategy for the last quarter of 2025/26, and

the successor had not been released as at July 2026. Elsewhere the pattern repeats: the police began digitising case dockets with a private partner in April 2025 at five stations nationally, while the mandated school administration platform, SA-SAMS, is reported at near-universal adoption at school level.

The centre is better served than the line. A financial management information system covering treasury, budget and further modules runs at central government level on a hybrid of custom and commercial software, payroll on custom software, and a public-service human resource system keyed to the national identification number. A treasury single account is used extensively by all departments and agencies. The system meant to unify the financial estate has slowed: the integrated financial management system mandated by Cabinet in 2005 had cost R2.6bn by September 2024, with its 2016 commercial licences to be converted to cloud, while the legacy accounting system's cutover to a new chart of accounts on 1 April 2025 strained its mainframe and left 29 of 162 departments forced to close April 2025's books. The customs system interoperates with other government systems; the social insurance system does not. Tax administration runs on commercial off-the-shelf software, and the revenue service ran the 2026 filing season with auto-assessments from 1 to 12 July while scaling artificial intelligence and analytics to profile taxpayer risk.

The police told Parliament in May 2021 that the electronic docket system was live at 998 stations, with 159 waiting on network upgrades. The 2025-2030 strategic plan commits to full docket integration and blames the state IT agency for delays. A ministerial reply of December 2025 records victim-data errors halting the women-and-children crime statistics. The courts run their own platform: Court Online, the judiciary's e-filing and evidence system, was live in 19 courts by August 2025.

Other GovTech and e-Gov

Three days before the 7 August 2026 deadline the Electoral Commission put voter registration inside WhatsApp, the whole flow — authentication, identity-document upload, address verification, voting station — in the chat, and 1.7 million transactions ran through it over the final weekend. The commission's own portal was already carrying the bulk of the work — 74% of more than 815,000 registration transactions since June 2026 — while its online candidate-nomination system had been used by 76 parties and 39 independent candidates, against 620 registered parties.

The unified front door is still being built. MyMzansi, the single national portal meant to replace an estate of separate departmental sites and to authenticate citizens by digital identity, moved from proof of concept to production-ready pilot by July 2026. Its first credential is the digital matric certificate, with September 2026 the target for hardening the pilot into a production platform and integrating it with the data exchange. The home affairs minister opened a digital identity credential in the MyMzansi app in a live demonstration on 1 October 2026, and gave no date for public release. The Presidency's Digital Services Unit built the prototype in about ten weeks on an open digital-public-goods stack, bundling a digital identity, a digital driving permit and an offline-capable wallet.

Home Affairs has shipped faster at the border than at home. The Electronic Travel Authorisation, now the flagship of its digital programme, went live nationally at OR Tambo in August 2026, extending to all qualifying countries from September and all visa types by year-end, running machine-learning document checks, facial recognition and rules-based automated decisions with no human review; the department said in October 2026 that a human is brought in where the system detects an anomaly. It is live at four airports — OR Tambo, Cape Town, King Shaka and Lanseria — from 12 August 2026, with 6,126 fraudulent applications rejected, put at 2.8% of pilot submissions. The department publishes no statistical series, and the four rejection counts given during 2026 rest on bases that do not reconcile.

The older estate works. eHomeAffairs lets citizens authenticate with an identity number and a one-time password and apply for Smart IDs and passports, eFiling is the main channel for filing and managing tax, and the national service portal is in use at transactional maturity, meaning services can be started and completed online; South Africa ranked 40th of 193 on the UN E-Government Development Index in 2024, entering its "very high" group for the first time. Against that, Operation Vulindlela records that modernisation of the Gov.za portal was unblocked only once the institutional arrangements delaying it were resolved, and concedes progress has been uneven and in places slower than anticipated. The state IT agency moved from a qualified audit opinion on 2024/25 to an unqualified one for 2025/26, on the minister's reply to Parliament. The electoral commission opened its results system to audit by experts designated by four parties, after an independent external audit (2026). The health department issued a bid for three years' hosting and support of the district health information system, run as nine provincial instances. The Auditor-General told Parliament the integrated justice system programme has not achieved its vision after about R4.9bn spent over a decade, with 63 of 506 planned network upgrades completed (2026). The social development department set 15 October 2026 for switching off its legacy registration and reporting system and ending over-the-counter submissions, leaving its online compliance system, introduced in September 2025, as the only route.

Home Affairs is to introduce an online, biometric intention-registration and appointment system for asylum applicants by 10 November 2026, announced in October 2026 and not yet operating.

Digitalisation

Digitalisation of sub-national government

South Africa's digital state stops being obligatory at the municipal boundary: the Minimum Interoperability Standards that national and provincial departments must comply with under the Public Service Regulations are only informative for local government, so the tier running most of the systems residents touch is under no duty to make them exchange data with anything.

What that tier builds, it buys piecemeal. ICT tenders advertised in a single week of July 2026 locate the state's build priorities below the centre: a five-year SITA server maintenance contract, a provincial VPN security upgrade for the Free State Office of the Premier, a unified channel-management contact centre for the City of Cape Town spanning emergency and non-emergency services, district-office server infrastructure for the Eastern Cape agriculture department, and DBSA consulting to turn the DCDT and Post Office business-rescue enquiry into a costed delivery route.

Johannesburg shows what the missing obligation looks like in service. The City's citizen call centre — the main route for residents to report outages, leaks, spillages, potholes and signal failures — went down in July 2026, the City saying only that services were never suspended and the downtime temporary; the same month it was under an Auditor-General regression finding on its audit outcomes and a National Treasury suspension of part of its funding. Its own 2025/26 draft integrated development plan concedes fragmented initiatives and insufficient governance structures, an admission published alongside a smart-city announcement for the City. The City of Tshwane and the CSIR launched an Enterprise Data and Analytics Platform on 29 September 2026, pooling municipal departments' service-delivery data after a six-month test, while keeping the live service-delivery dashboard internal. Johannesburg's power utility put out to tender a near-real-time monitoring platform for prepaid electricity purchases in October 2026, against R3.96bn of non-technical losses in 2024/25, allowing AI anomaly detection only if explainable, auditable and POPIA-compliant.

Municipalities are nonetheless held to account as data controllers and act as planning authorities over digital infrastructure. The Information Regulator obtained High Court confirmation of a fine against Blouberg Local Municipality for unlawfully publishing an employee's personal information (April 2026), and Cape Town's planning tribunal created a dedicated land-use class for data centres rather than fitting them into an existing industrial category (July 2026). The Electronic Communications Amendment Bill would standardise municipal by-laws, including uniform wayleave processes for electronic communications facilities (2026) — a national fix for a municipal inconsistency. Municipal capability also sets the floor under everything else digital: with load shedding ended, most household power interruptions now originate in municipal distribution networks (2025).

A 2020 parliamentary reply named councils budgeting in spreadsheets outside their core financial systems. The Treasury's 2024/25 budget circular required reporting only from the core financial system. One metro's audited accounts disclose an October 2023 ransomware attack that took down its financial system. The Treasury's 2026 report finds councils still reporting from spreadsheets.

Rural digital data capture

Power is no longer what limits recording data outside South Africa's cities: rural electricity access has overtaken urban access (2023), the product of decades of grid and off-grid extension set against backlogs in urban informal settlements, with a subsidised solar home system route for deep rural households where the grid does not reach. What still stops capture happening at the point of contact is connectivity, and the least-connected municipalities cluster in Mpumalanga, the Northern Cape and KwaZulu-Natal (March 2026). Rural primary healthcare clinics remain paper-based at the point of care, clinicians writing encounters by hand and entering them into the national health systems only afterwards (2025); rural school administration follows the same pattern (2025), and case-docket work in rural police stations still runs substantially on paper (2025). The digital record exists; it is assembled at a desk, later, from paper written in the field.

The policy answer runs through spectrum rather than through capture systems. The 2026 National Radio Frequency Plan, approved in July 2026, adds licence-exempt spectrum for Wi-Fi in rural and underserved areas — the minister's stated route for smaller providers priced out of licensed spectrum — and ICASA is launching a digital Electronic Frequency Information System portal alongside it. Spectrum is also the community network sector's single common demand of the regulator, which the Electronic Communications Amendment Bill answers with a sharing mandate on secondary licensees rather than an assignment of spectrum (July 2026).

Registration in the field stays an exception rather than a standard service. Online birth registration sits in the maternity units of larger hospitals rather than rural facilities (2025), and mobile units are the main civil-registration outreach to remote communities (2025). Where SASSA's facial biometric check fails on the applicant's side — poor lighting, an unstable network — the remedy on offer is to travel to a local office and enrol by fingerprint (May 2026). Rural South Africa is thinly measured from above as well: the Ibrahim Index leaves the country unscored on all four of its rural economy indicators (2023), so it cannot be placed on that measure at all.

Technology

AI

South Africa's banking supervisor has written artificial intelligence into its rulebook while the national policy remains unwritten: the Prudential Authority's 2025/26 report builds AI, cloud, cyber and data-offshoring into bank supervision and adds new cyber and cloud standards, reporting a first nationwide AI-adoption survey whose named barriers are regulatory uncertainty, AI skills and governance, as the five largest banks take AI out of pilots and into fraud detection, risk management and customer service (July 2026). No national AI policy framework is gazetted, the

April 2026 draft having been withdrawn rather than finalised, and the redraft is being prepared against a parliamentary committee whose chair is set against the new AI bodies the draft proposed — a regulatory authority, an ethics board, a commission and an ombud — arguing that the regulators already in place should be equipped for AI instead (July 2026). South Africa has meanwhile signed the agreement establishing the China-led World AI Cooperation Organization, set up in Shanghai in mid-July 2026.

Deployment is well ahead of the instrument. SARS applies machine learning to tax compliance, using it both to identify undeclared income and to determine which taxpayers are assessed automatically (July 2026), and its AI assistant, Lwazi, answered more than 2.7 million taxpayer questions in 2025/26; Home Affairs runs automated risk scoring over travel authorisation applications without a published appeal route; and the courts have begun setting the standards for AI-assisted legal work themselves, ahead of any statute. In health the regulator says plainly that it has not caught up: SAHPRA states that formal regulatory pathways for AI and machine-learning enabled medical devices are yet to be developed, naming its 2025-2030 strategic plan as the vehicle for closing the gap, and in the interim there is no route for systems that change after approval — the algorithm version is locked at licensing, significant changes must be notified, and the Authority is waiting on harmonised international guidance (September 2025). Among South African SMEs already using AI, 61% have deployed data analytics and customer-facing applications while most use cases have not reached deployment at all (February 2026), so the rules now being drafted are being drafted about systems that are mostly not yet in the market. Insurers are writing their own AI frameworks while the national policy is redrafted, using AI in claims, underwriting and fraud detection (October 2026).

ICT Industry

South Africa's largest fibre wholesaler now competes with its own customers: Openserve launched its own internet service provider in July 2026, putting it into retail competition with the ISPs it sells capacity to. Its parent's mix shows what the domestic sector is actually selling: group data revenue rose 8.8% to R6.9bn in the quarter to 30 June 2026 and took 62.4% of total revenue against 58.8% a year earlier, on group revenue up only 2.6%, so data's share is rising compositionally rather than through growth, while the group's IT services arm shrank over the same quarter with security and cloud the only lines growing.

Domestic firms hold particular layers rather than the stack. A Cape Town biometrics vendor is moving from one-off remote onboarding checks to continuous identity verification, with no South African rule allocating who should operate such a service (July 2026); a Midrand engineering business supplies the power and cooling plant under the foreign-owned data centre build. Consolidation in merchant commerce is domestic too: Pepkor is merging its Flash business with Shop2Shop into a single merchant-commerce platform, announced in July 2026 and still subject to regulatory and competition approval. The startup layer feeding this is funded at a scale the Finance chapter sets out, and the state records a startup and innovation support framework with financing and small-business support in place (2025).

Where the state buys rather than builds, the constraint is its own procurement: central ICT procurement through SITA is a national bottleneck, with contracts stalling for long periods and departments seeking exemption from using it (July 2026). Absorption on the private side is uneven — in mining, executives rate their own AI readiness and data management poorly and most have deployed nothing (2026). The structural position underneath remains strong: economic diversification scores 91.1 out of 100 on the 2024 Ibrahim Index, fourth of 54 African states, though it has fallen 6.7 points since 2014.

Innovation ecosystem

South Africa led the continent in both equity funding and equity deal count in 2025 for the first time since 2017, on sustained deal flow across stages rather than a handful of outsized rounds, with a single megadeal accounting for 15% of its funding. Against that, it ranked 69th of 133 economies on the WIPO Global Innovation Index 2024 — money is reaching companies faster than the wider system converts research into output.

Inside the state the machinery exists but its output does not surface: a public-sector innovation strategy or programme is recorded and a dedicated innovation lab established, while the results of that work are held internally rather than published (2025), which leaves nothing for another department to reuse. The private hub network runs across the main centres, with member hubs in Johannesburg, Cape Town, Durban and Stellenbosch (2025). Absorption by the largest institutions is slower than the funding suggests: the Reserve Bank's distributed-ledger work remains confined to two experiments that never moved into production.

Capacity

Literacy

South Africa gets almost every child through school and teaches them unequally: Education Completion scores 94.7 out of 100, fourth of 54 African countries, while Equality in Education stands at 44.9, thirty-second of 54 and effectively unchanged across the decade to 2023. The aggregate flatters that split — education overall scores 62.0, tenth of 54 and up two points over the same decade, with quality the fastest-moving component at 50.7 but still only seventeenth. The same profile records no score at all for Human Resources in Education, one of five indicators for which South Africa supplies no data, so the teaching workforce is the part of the picture that cannot be read.

Whether people can use a digital service is now being argued in court. In litigation over the SRD grant the state's position is that complaints applicants post on social platforms prove online access is no barrier, against applicants who say the system is an insurmountable hurdle for those without digital literacy or with disabilities (January 2026). The conditions behind that argument are physical as much as personal: a large share of public schools sits far from the nearest fibre (March 2026), and women remain markedly less likely than men to use mobile internet (2024).

For the public service, the National School of Government trained 112,855 learners in 2024/25 and signed an agreement with Microsoft to train 100,000 public servants in digital literacy, AI and cybersecurity (April 2025).

Training and skills

The weakest point in South African public-sector cyber resilience is staff rather than systems: in the CSIR's 2024 survey of 291 institutions, 7% had trained no employees at all and 32% had trained a quarter of their staff or fewer, with just 14% training more than three quarters. The shortage is also a hiring one — 62% of cyber security roles are partially or fully unfilled (2023).

The state has the apparatus. A public-service digital skills strategy and a programme delivering against it run from basic digital literacy up to advanced data science, and digital skills programmes are available at no cost (2025). Above them sits the National Digital and Future Skills Strategy, gazetted in 2020, which spans foundational, intermediate and highly specialised skills alongside technology hubs, regulatory sandboxes and artificial intelligence; the department told parliament in July 2026 that it is under review and a National Competency Framework is in development.

That framework carries a heavy load. The pipeline it has to mend is constrained at both ends on a development bank's review with the higher education department, which puts about 7 per cent of ICT and engineering posts unfilled and names software developers, network engineers and ICT security specialists as the hardest-to-fill roles,

and finds employers attributing the gap 26 per cent to shortfalls in basic-level training, 24 per cent to graduate pipeline throughput and 21 per cent to the pace of technological change (June 2026) — a mismatch the review holds to be structural and not self-correcting. Asked what artificial intelligence would do to South African employment, the department declined to forecast a net effect at all, holding that the outcome will depend on the country's ability to equip workers (July 2026) — displacement recast as a training problem rather than a labour-market one. Employers frame it the same way, with mining executives naming digital skills rather than technology as the primary barrier to adoption (2026).

Much of the money behind the training is not the state's. Germany funds a second phase of its Digital Skills for Jobs and Income programme to 2027, Microsoft's expansion pairs data-centre investment with AI skills training (April 2026), MTN's investment-conference pledge to 2028 covers digital skills alongside network build, and Google has committed to a digital innovation centre at a Soweto TVET college (2026). An industry-led digital skills impact fund had mobilised R600m and reached 4,000 young people by October 2026, and its backers are asking for R2.5bn more from existing state skills funds. Against those headline commitments, one small programme trained a named cohort for a named purpose: the South Africa School for Digital Policy's community networks edition trained 23 practitioners from community operators specifically to file submissions on communications regulation (2026).

Research institutions

The country's national research computing centre has run out of room: the Centre for High Performance Computing has reached maximum capacity and cannot accommodate future growth, as parliament's science committee recorded on an oversight visit in August 2026 while attaching no budget, timeline or capacity target to the sustained investment it called for. It sits inside the state's own National Integrated Cyber Infrastructure System, run by the CSIR for the science department. Age is the other half of the problem, and it has already been exploited: the science minister told the National Assembly that the late-May 2026 breach of the Lengau supercomputer was caused by vulnerabilities associated with the legacy high-performance computing system, that the intrusion reached parts of the environment and deployed crypto-currency mining malware, and that ageing components are technically constrained or no longer fully supported by their original vendors. A remediation programme covering access controls, privileged-access management, network segmentation, patch management and monitoring has since been run, with no cost, data-loss finding or outage duration stated. Where research capacity is protected, it is protected by spectrum rather than money — the National Radio Frequency Plan 2026 shields the radio-astronomy reserve around the Square Kilometre Array.

The University of Cape Town and the Global Center on AI Governance launched the African Hub on AI Safety, Peace and Security on 28 September 2025, under the AI4D programme and financed by the UK's FCDO and Canada's IDRC, to carry African priorities into a global AI-safety debate set elsewhere.

Inclusion

Access to services

A social grant flagged by South Africa's verification checks — by a small unemployment-insurance payment, by bank-account income data, or by a stale employment record — is suspended with no effective correction route, leaving beneficiaries flagged in September 2025 still unpaid (July 2026). The burden of disproving the flag falls entirely on the beneficiary, who must produce payslips and bank statements and attend an office in person, with an income investigation running for months (July 2026).

That machinery sits on a caseload the Department of Social Development's own numbers put beyond reach: 13.4 million working-age people had no income at all and 18.3 million had incomes below the food poverty line (June 2022). When the SRD eligibility criteria were tightened, applications fell from over 15.8 million to just over 8.1 million and approvals from 10.9 million to 5.6 million between March and April 2022. The high court has since found SASSA's failure to pay successful applicants unconstitutional.

The physical counter carries what the digital channel cannot. SASSA's community registration outreach in the Western Cape had delivered 14 of 56 events planned for the year by the end of the first quarter (August 2026), while grant beneficiaries were being moved off gold payment cards due to stop working after 31 August 2026, and Home Affairs cannot trace the people whose identity numbers it wrongly blocked on the population register.

State connectivity is not closing the distance. SA Connect, the flagship broadband programme, was effectively frozen for most of FY2025/26: its in-year allocation collapsed from an expected R1.1bn to R116m, the money arrived only in January 2026, leaving the 100%-access phase-2 target set in 2022 well out of reach, though 7,397 public facilities were connected (June 2026). Only 10,851 of 21,878 public service institutions targeted under spectrum licence obligations were connected as at March 2026, 49.1%, with two operators below the 65% minimum. The justice department names digital exclusion as an open problem in its own digitisation of the courts, identifying rural communities, the elderly and the poor as the groups at risk (July 2026). On the Ibrahim Index's measure of how equally public services reach social groups, South Africa scores 51.5 out of 100, up 12.1 points in a decade and still 18th of 54 (2023). About R652m of the universal service fund was unspent at the end of 2025/26, of R1.29bn collected over five years, on the minister's reply to Parliament, and the regulator's inquiry into the affordability of telecommunications was withdrawn on 30 September 2026, 26 days after it was gazetted, to be republished with a draft questionnaire.

The social development department received a qualified audit opinion for 2025/26 over weaknesses in the agency's payment system, including grants paid to deceased people and government employees, Parliament's committee heard on 6 October 2026. The agency defended its reviews and biometric enrolment as fraud controls on 9 October.

Digital divides

Exclusion in South Africa is now most measurably produced by the state's own verification: 67,868 social grants were suspended and 48,949 lapsed over the first three quarters of FY2025/26 for failure to complete review or verification, overwhelmingly Child Support Grants — 37,825 of the suspensions and 42,748 of the lapses — after income testing of some 6 million Child Support Grant recipients was run with the banks (March 2026).

The design that produces those numbers has been through court. In January 2025 the North Gauteng High Court declared the regulation confining COVID-19 SRD applications to the online channel unconstitutional and invalid, curing it by reading in the words "or at the offices of the Agency"; the state took the ruling on appeal, and the Supreme Court of Appeal heard it on 25 August 2026 and reserved judgment pending further affidavits, having admitted an artificial-intelligence governance organisation as a friend of the court. The Department of Social Development's position in that litigation is that complaints posted by applicants on platforms such as X prove there is no barrier to online access.

Behind that argument, 13.6 million people were offline at the start of 2025, 21.1% of the population, against 50.8 million internet users. The gradient is gendered: women are 14% less likely than men to use mobile internet (2024). It is also priced — South Africa ranked 31st of 45 African countries in the second quarter of 2025 for a monthly 1GB mobile package, even as wired and wireless prices tracked below headline inflation between April 2025 and January

2026. It clusters geographically in Mpumalanga, the Northern Cape and KwaZulu-Natal, and it reaches into the counter itself: an applicant whose facial biometric fails on SASSA's online portal, for poor lighting or an unstable connection, is told to travel to an office and enrol by fingerprint.

One channel runs the other way. Women account for 61% of online voter registration transactions and 16–29-year-olds for 50%; just over half the young cohort's transactions were new registrations, and since the June 2026 registration weekend about 603,000 of more than 815,000 transactions have run through the Electoral Commission's data-free portal (August 2026).

Data

National statistics

South Africa has conducted no business or establishment survey in the past ten years, the one collection missing from a statistical programme the World Bank otherwise scores as complete (2023): a population census, an agricultural census and a business census all fall inside the same window, as do three or more rounds each of household, labour force, health and agricultural surveys.

The programme is strong by both measures that grade it. South Africa sits in the 70-89% band on all five of the World Bank's pillars — data use, services, products, sources and infrastructure (2023) — and the Ibrahim Index puts the capacity of the statistical system at 72.7 out of 100, fifth in Africa and 1.6 points up over the decade to 2023.

The weakness sits at the edges rather than in the statistical office's core work. Several Ibrahim indicators carry no South African score at all, including the whole rural economy sub-category and the measure of staffing in education, for which no data is supplied. Reporting has in places moved away from series that can be tracked: Home Affairs replaced its periodic count of participating Smart ID bank branches with a live locator in August 2026, and the Electronic Travel Authorisation's usage reaches the public through speeches rather than any published series (August 2026). The National Cybersecurity Hub, for its part, publishes no statistics of any kind on the incidents it handles.

Statistics South Africa was itself hit by a ransomware attack in March 2026.

A deputy minister in the Presidency said in a keynote address on 9 October 2026 that Census 2031 is in development, and called statistical independence a constitutional safeguard.

Open data

South Africa ranks first of 54 African states on both of the Ibrahim Index's records measures — disclosure of public records at 80.0 out of 100, up 12.4 points since 2014, and accessibility at 75.3, up 3.5 (2023) — and yet the data describing public services is the least open it holds, with health facility data scoring between 0.5 and 2.5 out of 10 and education facilities between 3 and 5 in the 2024 Open Data Inventory.

The pattern runs through the whole inventory. Government finance, poverty and crime each score between 3 and 5 out of 10 (2024), below the macroeconomic series where the country does best: between 5.5 and 7.5 for labour, national accounts, balance of payments, trade, prices, population and agriculture. Numbers describing the economy are published well; numbers describing the state's own delivery are not.

A national open data portal is running, publishing basic information and datasets on a mix of annual, quarterly and monthly refresh cycles (2025), and a data governance body exists without any published account of how data across government systems is governed (2025). The right-to-information law is in force and rated effective, though the

same index marks the country down on whether the public can obtain records in practice or ask for what is not already published, and South Africans reach data held about them through statutory request routes rather than any self-service facility.

Where the data is infrastructural, it stays private. The dataset behind South Africa's first broadband map is owned by the Development Bank of Southern Africa and has not been released, and the study built on it criticises government and ICASA for withholding infrastructure data (March 2026) — the same regulator that has since moved to compel operators to file geo-referenced infrastructure data. The government data exchange runs with no live public dashboard, public access to aggregated datasets being planned rather than delivered (2025). Enforcement, meanwhile, has reached private bodies: the Information Regulator ordered a mining house to disclose its social and labour plan reports in June 2026.

Use of satellite data

The state put its own name to a satellite operator in July 2026, when Deputy President Paul Mashatile launched BrainSAT Satellite Services — a South African company running on Emirati Space42 capacity, Ka-band broadband and Thuraya handsets, under a national SATCOM strategy whose stated destination is an eventually South African-owned satellite. Amazon Leo will enter in 2027 through Herotel as its licensed local partner, an arrangement that raises no ownership question. Starlink has neither: it reached its 27th African country in July 2026 without launching in South Africa, its rollout held up on black economic empowerment compliance, and the only lawful route left to a new entrant is acquiring an existing licensee under the Electronic Communications Act.

On the data itself, the state's standing national geospatial holding is the Chief Surveyor General's digital cadastral layer of farms, erven and holdings across all provinces. Operational use runs through the Weather Service, which receives EUMETSAT satellite imagery every 15 minutes alongside 14 radars and 214 automatic weather stations (November 2024).

Geopolitics

US / hyperscaler activities

The size of the American position only became public in March 2025, when Microsoft disclosed R20.4 billion of spending over the preceding three years on the Johannesburg and Cape Town facilities behind the Azure regions that opened in March 2019, the first enterprise-grade cloud regions on the continent. Amazon Web Services put R15.6 billion into its Cape Town region between 2018 and 2022, a region launched in April 2020 with three availability zones, and Equinix entered in December 2022 with a $160 million carrier-neutral colocation and interconnection data centre in Johannesburg, since opened. Google's Johannesburg cloud region, the first it has built in Africa, followed in March 2025. Between them these four hold the floor the country's cloud market now runs on.

American public money is an order smaller and sits behind private operators rather than the state. The US DFC disbursed $83 million to Africa Data Centres in August 2022, primarily for its South African expansion, and its predecessor OPIC had in 2015 guaranteed up to $80 million to IHS Fund II SA, a fund investing in telecommunications tower infrastructure alongside other development lenders. USTDA has funded five ICT and connectivity feasibility studies and technical assistance grants between 2017 and 2023 — rural TV white space Wi-Fi, Dark Fibre Africa's peri-urban expansion, Western Cape broadband and off-grid township connectivity among them — none larger than about $2 million.

Where an American operator has pushed on the rules rather than worked around them, it has not got in. A December 2025 policy direction letting foreign operators substitute equity-equivalence investment for direct Black ownership was read as opening the door for Starlink, and ICASA rejected it in May 2026. Amazon's Leo service is instead due in

2027 through a licensed local distributor, an arrangement that raises no ownership question at all. The dependency deepens at the edges too: Africa Data Centres' Johannesburg site became an Azure ExpressRoute Metro peering location in July 2026, hardening the private route into a US cloud, and voter registration now runs in part through a US-owned messaging platform.

China activities

South Africa has signed the agreement establishing the China-led World AI Cooperation Organization, set up in Shanghai in July 2026. On the ground the Chinese position is older and narrower than that alignment suggests, and it is shaped by vendors rather than by the state: Chinese money in the digital record arrives as credit tied to equipment purchases, not as policy-bank lending to government systems. Cell C took a capital expenditure facility from the Industrial and Commercial Bank of China in 2015 of $240 million plus R1.3 billion, financing a second frame supply agreement with Huawei for radio access equipment and the expansion of its LTE, mobile broadband and microwave networks, half-guaranteed by Oger Telecom and maturing in June 2023. Telkom raised a ten-year syndicated buyer's credit from ICBC and Standard Bank in 2019, insured by Sinosure, to buy 4G equipment from Huawei — the first rand funding it had raised from China.

Huawei is the one Chinese party holding infrastructure rather than paper. It has operated cloud availability zones in the country since 2019, the first international provider with a hyperscale data centre in Africa, and opened a third Johannesburg zone in January 2023. Standard Bank Group expanded its partnership with Huawei in September 2026 to move its core banking systems onto a hybrid cloud in phases; no value, hosting location or data-residency terms are published.

EU activities

European money in South Africa's digital economy buys skills and minority equity rather than infrastructure. Germany's BMZ funded Digital Skills for Jobs and Income with a EUR 9.6 million grant running from 2020 to 2024, aimed at the employment prospects of young people and of women in particular, and is funding a second phase with EUR 6.5 million running from 2024 to 2027. Finnfund has invested EUR 2 million in Fibertime Group (2023), which sells low-cost fibre in township communities that conventional rollout has not reached.

The rest is venture equity in financial technology, taken beside commercial investors rather than in place of them. Germany's DEG put $4 million into insurtech Naked Insurance's $38 million Series B2 round in 2025, led by BlueOrchard with Hollard, Yellowwoods and IFC continuing, and took a place in the 2023 Series B for Lulalend, the country's leading digital lender to small and medium enterprises, a round led by Lightrock alongside Triodos, Women's World Banking, IFC and Quona.

Gulf/UAE activities

The Gulf's foothold in South African connectivity is a state-backed national operator running on Emirati satellites. Deputy President Paul Mashatile launched BrainSAT Satellite Services in July 2026, a South African company selling YahClick Ka-band broadband and Thuraya 4 handsets over infrastructure owned by the UAE's Space42, under a national SATCOM strategy whose stated destination is an eventual South African-owned satellite system. The Department of Science, Technology and Innovation makes the case for that system as import substitution, setting what the country spends each year with foreign communications providers against the one-time cost of a locally owned satellite with a twenty-year life. Until it exists, the capacity the state has just put its name to is Abu Dhabi's.

India activities

India held no financing, infrastructure or platform position in South Africa's digital economy as at August 2026.

Digital sovereignty

Home Affairs' plan for a state digital identity lists a certificate authority among the things it will build, without saying who would hold the keys, while South Africa's existing trust layer for electronic signatures is delegated to three accredited commercial operators (September 2026).

The rule on where state data sits is policy. The National Policy on Data and Cloud of May 2024 makes SITA the authority that sources data infrastructure and cloud services for government from industry, and requires government data to be stored in unified cloud-enabled data centres; the communications department told Parliament in February 2026 that government data bearing on national security and sovereignty is to be stored only on infrastructure inside the country, and that it wants a national integrated government data centre run by a repurposed SITA. The President wrote in July 2026 that government is investing in its own cloud through the CSIR, citing countries where sensitive data has been held by private firms outside national jurisdiction.

Owning a system has proved separate from paying for it. The skills authority financing a learner-management system for the whole post-school sector was advised that the university which built it owns the intellectual property, and is seeking co-ownership; it hosts the system at the Dube Tradeport data centre and has warned that the lack of government-owned data centres exposes national data (August 2026). Whether the state holds the source code and operating skills for the national identity system, or can exit its supplier contracts, was not on record as at September 2026. The terms on which the police service's Huawei cloud contract and the hyperscaler services SITA brokers to departments govern data jurisdiction, portability and exit had not been published as at September 2026.

The agency meant to hold all this has set itself a test. SITA's board chair told GovTech 2026 that sovereignty means knowing where data is stored, who may access it and how services continue when a supplier fails, and that management will report against verified targets for service availability, cost, resilience and local economic participation by GovTech 2027 (September 2026).

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